

CONTACT

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ORCID 0000-0001-7758-695X
PROJECT graphole.com
LOCATION São Paulo, Brazil
RELOCATION Open to relocation (on-site)

SKILLS

Languages

`Java` `Python` `Go` `Kotlin` `C` `C++` `JavaScript` `SQL`

Machine Learning & AI

`Deep Learning` `Neural Networks` `Computer Vision` `NLP` `PCA` `SVM` `Clustering`

Backend & Architecture

`Spring Boot` `Spring Cloud` `Microservices` `Design Patterns` `Software Architecture`

Data & Infrastructure

`PostgreSQL` `Redis` `Kafka` `RabbitMQ` `Docker` `Kubernetes` `Spark` `MongoDB` `Neo4j`

Domains

`Banking & Fintech` `Cryptography (PKI/HSM)` `High-Performance Computing` `POS/ERP`

EDUCATION

D.Sc., Electrical Engineering
Polytechnic School, USP 2007–2012

M.Sc., Electrical Engineering
Polytechnic School, USP 2003–2006

B.Sc., Computer Science
FEI 1999–2002

LANGUAGES

Portuguese	Native
English	Advanced
German	B1

Humberto Sandmann

Senior Software Architect & AI Engineer

PROFILE

Senior software architect and artificial intelligence engineer with 20+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, AI, and distributed systems.

EXPERIENCE

Founder · H-Sandmann 2006–2025

Independent consultancy delivering short-term, specialized engagements in machine learning, optimization and security.

`Machine Learning` `Software Architecture`

Co-Founder · SelsanTech 2012–2023

Co-founded a venture bringing computational-intelligence research to market; built ZigTrack, real-time indoor tracking of equipment with low-cost fixed sensors.

`Machine Learning` `Software Architecture`

Senior Software Architect & Machine Learning Engineer · RockSpoon 2018–2019

Team leader and product owner; architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

`POS` `ERP` `JavaEE` `PostgreSQL` `OpenCV` `Computer Vision` `Go` `Android` `Kotlin`

Senior Consultant, Computational Optimization · Banco Votorantim 2015–2016

Return-on-equity computation for 2M+ records within a 12-hour window on high-performance computing infrastructure for a major bank.

`RoE` `High-Performance Computing` `J2EE` `Spring Boot` `Jenkins` `Test Automation`

Senior Consultant, Computational Optimization · TecBan 2011

Senior Software Engineer / Architect · ABN Amro Bank 2006–2007

Senior System Analyst · Politec 2005–2006

Senior Software Engineer / Architect · EverSystems 2004–2005

Senior System Analyst · Central Online 2002–2003

Senior System Analyst · Datamace 1999–2002

Computer Programmer · Heraeus Electro-Nite 1998–1999

ACADEMIC POSITIONS

Associate Professor · Insper 2018–present

Associate Professor · ESPM 2014–2025

Associate Professor (freelance) · Link School of Business 2024

Professor · IBTA — Instituto Brasileiro de Tecnologia Avançada 2016–2017

Associate Professor · FIAP 2012–2015

Visiting Professor · Toledo Prudente Centro Universitário 2015

Associate Professor · Universidade Cruzeiro do Sul 2012–2013

Temporary Professor · Universidade Nove de Julho 2011

Teaching Assistant · Polytechnic School, University of São Paulo 2008–2011

Full academic CV, publications and teaching record at hsandmann.github.io Updated October 2026